FIFA 19 Player Data: Data Cleaning & Visual Analysis

Project Proposal | Dataset: FIFA 19 Complete Player Dataset (Kaggle) | Tools: Python, Pandas, Matplotlib, Seaborn

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1. Project Definition

Football clubs and scouts now rely on data to judge players, set prices and plan transfers. This project cleans and analyses FIFA 19 player data to find out who the best players are, how age affects rating and value, and which clubs and countries hold the most talent. It delivers a cleaned dataset, seven charts, and a short insights report.

Problem statement

The raw dataset is large and messy. Money columns are stored as text (for example "€110.5M"), height and weight use mixed units, and many columns have missing values. This makes it hard to compare players or see patterns. A clean dataset and clear charts turn it into something a scout or manager can read quickly.

Objectives

- Clean the raw data so that every column has the right type and no hidden errors.
- Understand the spread of player age, rating, value and wage.
- Find the top countries and clubs by number of players and by average rating.
- Show how age affects rating, potential and market value.
- Compare wages across playing positions.
- Identify young players with the highest potential for scouting.

2. Dataset and Use Case

Dataset: FIFA 19 Complete Player Dataset on Kaggle: about 18,200 players and 89 columns, covering personal details, club, ratings, value, wage and skill attributes. The key columns are Overall, Potential, Age, Value and Wage.

Use case: football scouts and club managers can use the results to find young high-potential players, judge fair market value, and compare squads and countries.

Key columns used

Column	Type after cleaning	How it is used
Age	Integer	Age groups and age-based trends (charts 2 and 3)
Nationality	Category	Players per country (chart 1)
Club	Category	Average rating per club (chart 5)
Overall, Potential	Integer	Player quality and growth room (charts 3, 4, 5, 7)
Value, Wage	Float (euros)	Market value and pay analysis (charts 4 and 6)
Position	Category	Grouped into forward, midfielder, defender and goalkeeper (chart 6)

Data cleaning steps

- **Missing values:** fill Club with "No Club", fill missing skill attributes with the column median, and drop columns with mostly empty values (such as Loaned From).

- **Duplicates:** find and remove repeated rows, then confirm that every player ID appears once.
- **Data types:** convert Value, Wage and Release Clause from text such as "€110.5M" to numbers, convert Height and Weight to cm and kg, and convert Joined to datetime.
- **Outliers:** remove players with a Value of zero and extreme Wage values, using the IQR rule.
- **New features:** create age groups and position groups (forward, midfielder, defender, goalkeeper) to make grouping and charting easier.

Row and column counts will be printed before and after cleaning so that every change can be checked.

3. Visualizations and Outcomes

Only **line charts** and **bar charts** will be used. Line charts show change over a continuous measure such as age, and bar charts compare categories. Each chart gets a title, labelled axes and a one-line takeaway.

#	Chart type	What it shows	Expected outcome
1	Bar	Top 10 nationalities by number of players	England, Germany, Spain, France and Argentina lead
2	Bar	Number of players by age group (16-20, 21-25, 26-30, 31-35, 36+)	Most players are between 21 and 30 years old
3	Line	Average Overall and Potential rating by age	Overall rises until about 27 to 30 then declines; Potential is highest for young players
4	Line	Average market value by Overall rating	Value grows exponentially at higher ratings
5	Bar	Top 10 clubs by average Overall rating	Elite clubs such as Juventus, Napoli and Real Madrid rank highest
6	Bar	Average wage by position group (forward, midfielder, defender, goalkeeper)	Attacking players earn the highest average wages
7	Bar	Top 10 young players (under 21) by Potential	Highlights the best future stars for scouting

How each chart will be built

- **Chart 1:** count players per nationality, sort from high to low and plot the top 10 as a bar chart. It shows where football talent is concentrated.
- **Chart 2:** cut Age into five groups and count players in each. It shows which age groups dominate professional football.
- **Chart 3:** group by Age and plot the mean of Overall and Potential as two lines. The gap between the lines shows how much young players are still expected to grow.
- **Chart 4:** group by Overall and plot the mean Value. The curve shows how sharply prices rise for the best players.
- **Chart 5:** group by Club, keep clubs with enough players (for example 20 or more) so that averages are fair, and plot the top 10.
- **Chart 6:** group by position group and plot the mean Wage. It shows which roles are paid the most.
- **Chart 7:** filter players under 21, sort by Potential and plot the top 10. It gives a ready shortlist for scouts.

4. Python Libraries and Tools

Tool	Purpose
Pandas	Loading the CSV, cleaning, grouping and aggregating data

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NumPy	Numerical operations and the IQR outlier calculation
Matplotlib	Base library for line and bar charts
Seaborn	Cleaner chart styling and colour palettes
Jupyter Notebook or Google Colab	Running the code step by step with notes and charts together
Git and GitHub	Version control, hosting the code and documentation

5. GitHub Link and Documentation

Repository: https://github.com/mouryaVora03/fifa-19-player-data

The repository will contain the raw and cleaned data, the cleaning and visualization notebooks, exported charts, the insights report, requirements.txt, and a README explaining the project and how to run it.

Planned repository structure

- **README.md:** project overview, dataset link, objectives, chart screenshots, key findings and steps to run the code.
- **data/:** raw and cleaned CSV files.
- **notebooks/:** one notebook for cleaning and one for visualization.
- **charts/:** all seven charts exported as images.
- **report/:** the short insights report.
- **requirements.txt:** pandas, numpy, matplotlib, seaborn, jupyter.

6. Short Insights Report

After the charts are made, a short report will cover four points:

- **Data quality:** what was wrong with the raw data and how it was fixed, with row counts before and after cleaning.
- **Key findings:** the 4 to 5 most important patterns, each linked to one chart.
- **Recommendations:** for example, scout players under 23 with a large gap between Potential and Overall, expect steep price rises above an 80 rating, and compare clubs by squad average rather than star players alone.
- **Limitations:** the data covers a single game edition, ratings are game-designed rather than real match statistics, and the analysis describes patterns without proving causes.

7. Timeline

Step	Task	Time	Output
1	Explore the dataset and check columns	1 day	Notes on data issues
2	Clean the data	2 days	Cleaned CSV and cleaning notebook
3	Make the charts and write the findings	2 days	Seven charts and findings
4	Write the report and README, upload to GitHub	2 days	Report and public repository

Total time: about 7 days. Both team members will share the work: one leads cleaning and the other leads charts, and both review each other's code and write the report together.

8. Conclusion

This project covers the full analysis workflow on a real football dataset. The result will be a clean dataset, seven line and bar charts, a short insights report, and a GitHub repository that anyone can reproduce. It will give both team members practical experience in data cleaning, visualization and clear reporting with Python.